CYBER SECURITY TECHNICAL CLUB PROJECT

Phishing Detection Tool: Identifying & Preventing Fake Email

A Simple & Understandable C Programming Project for Beginners

Project Title:	PhishGuard (Fake Email Detector)	**Developed By:**	**Team APEX CIPHER**
Student Level:	1st Year B.Tech / CSE Students	**Language Used:**	C Language (stdio.h, string.h, ctype.h)

1. What is Phishing? (In Simple Words)

Phishing is a trick where a cyber criminal (hacker) sends a **fake email** pretending to be from a real trusted company like a Bank, Google, or PayPal. Their goal is to make people panic so that they click a fake link and give away their passwords, OTPs, or credit card numbers.



Figure 1: How an attacker sends a fake email and how our C program protects the user.

2. Why Did We Choose C Language for This Project?

In our first year of college, **C programming** is our fundamental subject. Building this project in C helped us practice:

- **Character Arrays (Strings):** Storing email sender, subject, and message.
- **String Functions:** Using strstr() to search for keywords and tolower() for uppercase/lowercase matching.
- **Conditionals (if-else):** Adding points whenever a suspicious pattern is discovered.
- **Cyber Security Awareness:** Learning real-world defensive techniques used in IT clubs and SOC teams.

3. The 4 Simple Checks Our Program Performs

Instead of complex formulas, our program performs **4 easy-to-understand checks** just like a human security analyst does:

Check 1: Fake / Lookalike Sender Email Address (+35 Points)

Attackers change small letters to trick our eyes (like paypa1.com with number '1' instead of letter 'l', or g00gle.com with zeros). If our code finds these fake spellings, it gives 35 risk points.

Check 2: Urgent / Panic Words in the Subject Line (+25 Points)

Fake emails always make people rush so they don't think carefully. They use words like: *"URGENT"*, *"Account Suspended"*, or *"Action Required"*. If found, our code adds 25 risk points.

Check 3: Dangerous Web Links & Raw IP Numbers (+25 Points)

Real companies never send raw computer IP numbers like http://192.168.1.50/login. Attackers use raw IPs or click here links to lead victims to phishing websites.

Check 4: Asking for Secret Confidential Information (+30 Points)

Real banks clearly state: *'We never ask for your password or PIN via email'*. If the email contains words like *'password'*, *'credit card'*, or *'OTP'*, it is almost certainly a scam (+30 points).

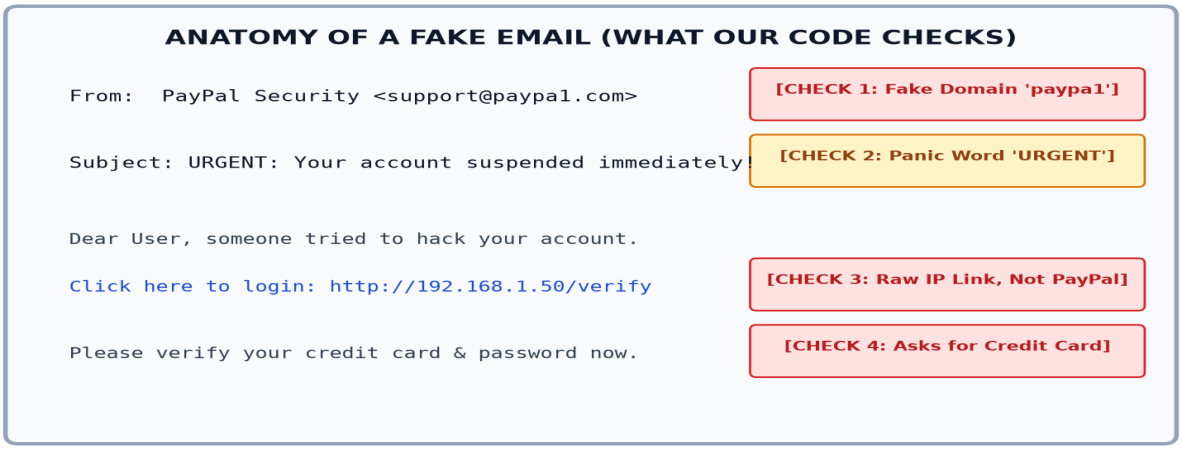


Figure 2: Example of a fake PayPal email showing the exact 4 parts inspected by our C code.

4. How the Risk Score is Calculated



Risk Score	Result Category	What the User Should Do
0 to 19 Points	**SAFE EMAIL**	The email looks normal and safe to read.
20 to 49 Points	**SUSPICIOUS**	Be careful! Don't open any attachments or links.
50 to 100 Points	**DANGEROUS PHISHING!**	High alert! Delete the email immediately.

5. The Beginner-Friendly C Source Code (`simple_phishguard.c`)

Here is the clean, easy-to-read C code written using simple 1st-year concepts (arrays, if statements, and strstr()):

```c
#include <stdio.h>
#include <string.h>
#include <ctype.h>

int main() {
    char sender[100], subject[150], body[500];
    int score = 0;

    // 1. Take input from user
    printf("Enter Sender Email: ");
    fgets(sender, sizeof(sender), stdin);

    // 2. CHECK 1: Fake / Lookalike Domain
    if (strstr(sender, "paypa1") != NULL || strstr(sender, ".xyz") != NULL) {
        printf("[!] Suspicious Sender detected! (+35 points)\n");
        score = score + 35;
    }

    // 3. CHECK 2: Urgent words in Subject
    if (strstr(subject, "urgent") != NULL || strstr(subject, "suspended") != NULL) {
        printf("[!] Panic/Urgent subject detected! (+25 points)\n");
        score = score + 25;
    }

    // 4. CHECK 3: Fake links with raw IP numbers
    if (strstr(body, "http://192.") != NULL || strstr(body, "click here") != NULL) {
        printf("[!] Dangerous link detected! (+25 points)\n");
        score = score + 25;
    }

    // 5. CHECK 4: Asking for password or credit card
    if (strstr(body, "password") != NULL || strstr(body, "credit card") != NULL) {
        printf("[!] Asking for sensitive card/password! (+30 points)\n");
        score = score + 30;
    }

    // 6. Show Final Result
    if (score >= 50) {
        printf("VERDICT: [DANGEROUS PHISHING EMAIL!]\n");
    } else {
        printf("VERDICT: [SAFE EMAIL]\n");
    }
    return 0;
}
```

Simple Functions Used in Our Code (Viva Q&A;):

- **fgets():** Takes string input safely from the keyboard without causing memory overflows.
- **strstr(str1, str2):** A standard C string function that checks if str2 exists inside str1. If found, it returns non-NULL.
- **tolower():** Converts letters to lowercase so capital letters like 'URGENT' or 'Urgent' both get detected.

6. Seminar Presentation Tips for APEX CIPHER

1. **Start with a Hook:** Show the fake email image and ask the class: *'Can you spot why paypa1.com is fake?'*
2. **Keep it Simple:** Explain the 4 rules clearly (Fake Sender, Urgent Subject, Raw IP Link, Asking Password).
3. **Run the Demo:** Type a fake email into the terminal and show the total score reaching 100 points!
4. **Conclusion:** Summarize that basic C programming logic is powerful enough to protect users from cyber scams.